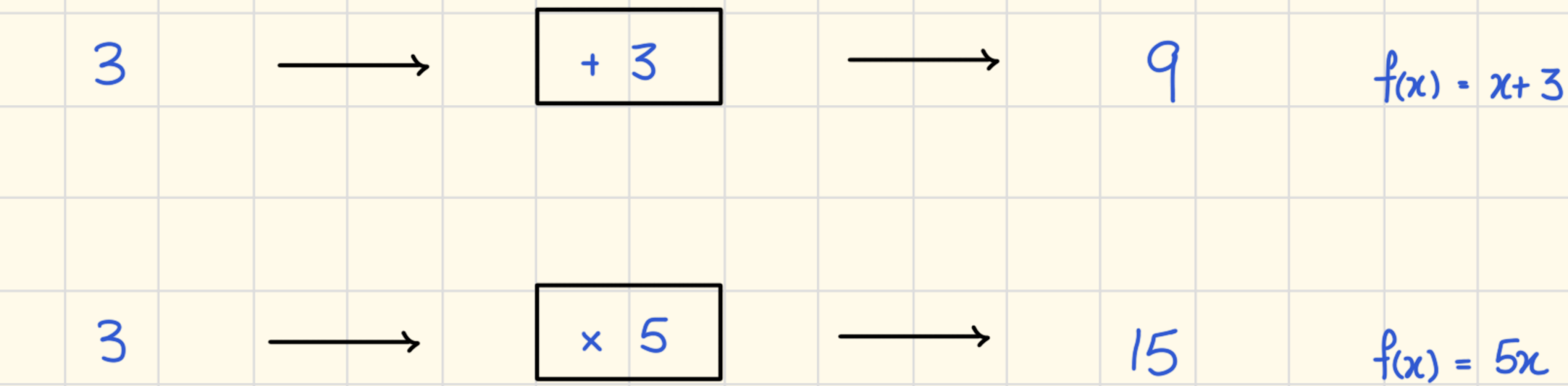
FUNCTIONS

A function is a machine that performs operations on input to get outputs.

input x → function f → output $f(x)$



Domain And Range

The domain is all the inputs (x) a function inputs.

The range is all the outputs (y) a function outputs.

Domain & Range

Set

$x = \{-1, 0, 1, 2\}$

$y = \{2, 3, 5\}$

Interval

$0 \leq x \leq 3$

$1 \leq y < 5$

Domain And Range

What is the range of $f(x)$?

Domain

$\{-1, 0, 1, 2\}$

Function

$f(x) = (2x+1)^2$

Range

$f(-1) = 1$, $f(1) = 9$

$f(0) = 1$, $f(2) = 25$

$y = \{1, 9, 25\}$

What is the domain of $f(x)$?

Range

$\{-1, 0, 1, 2\}$

Function

$f(x) = x + 5$

Domain

$x + 5 = -1$, $x + 5 = 1$

$x + 5 = 0$, $x + 5 = 2$

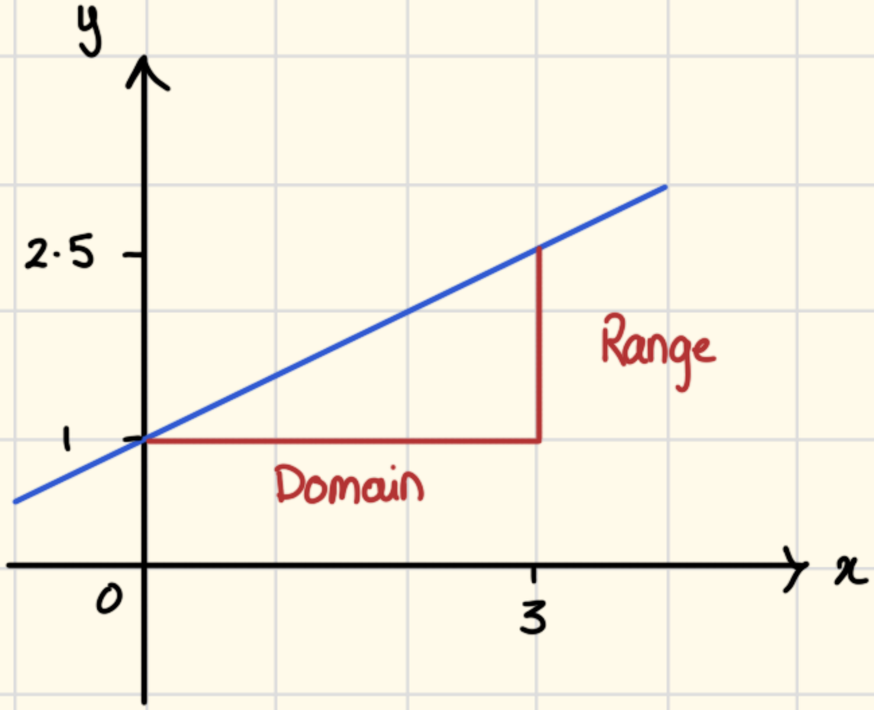
$x = \{-6, -5, -4, -3\}$

Domain And Range

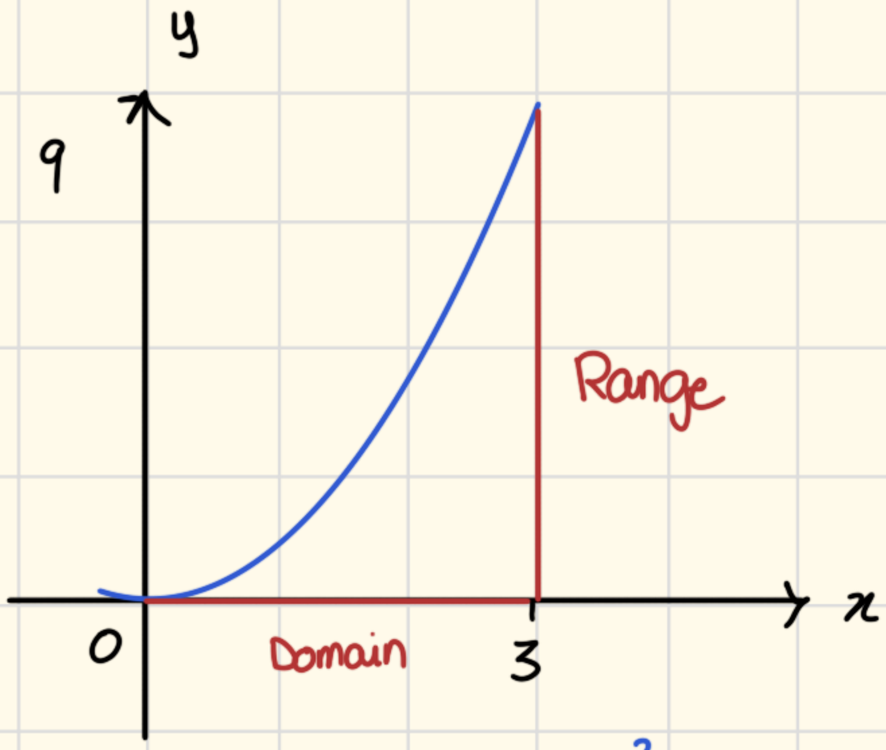
What is the range of function ?

Domain

$0 \leq x < 3$



$f(x) = \frac{1}{2}x + 1$



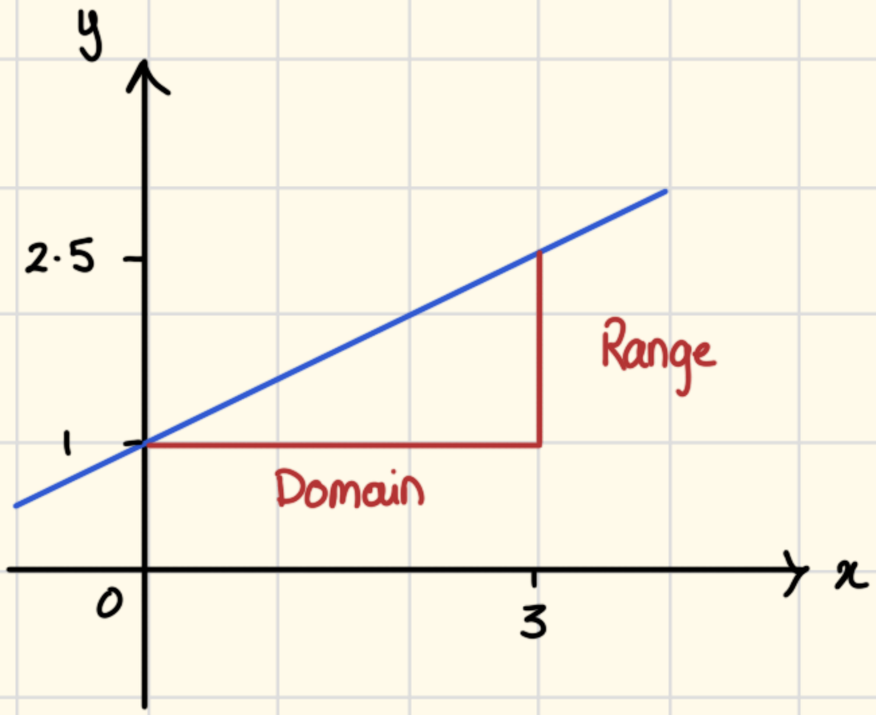
$g(x) = x^2$

Domain And Range

What is the range of function ?

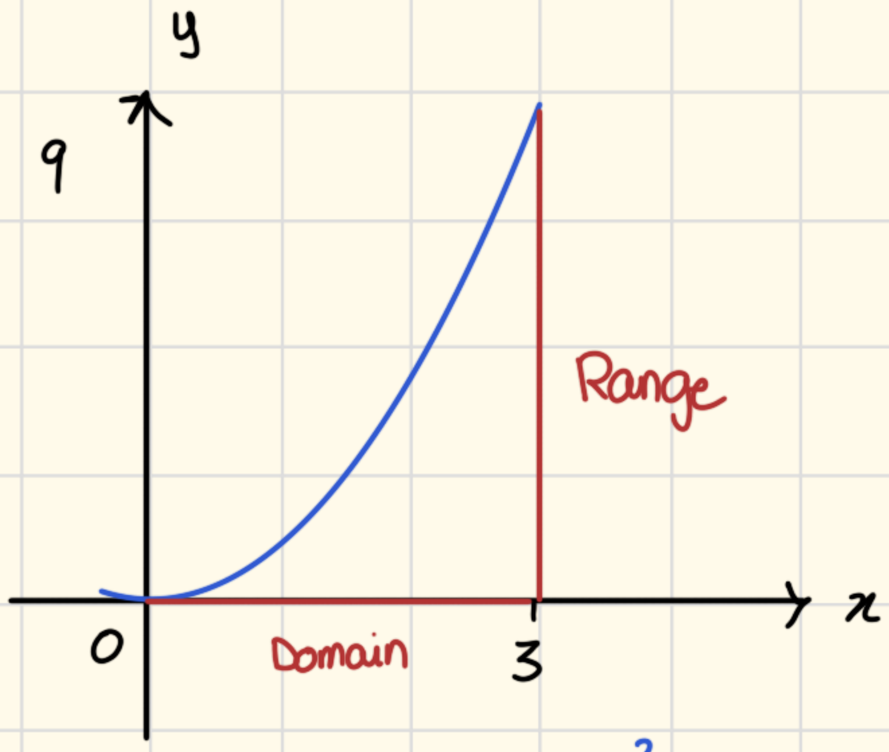
Domain

$0 \leq x < 3$



$f(x) = \frac{1}{2}x + 1$

Range & $1 \leq y < 2·5$



$g(x) = x^2$

$0 \leq y < 9$

Domain And Range

Special function can't take specific domain values !

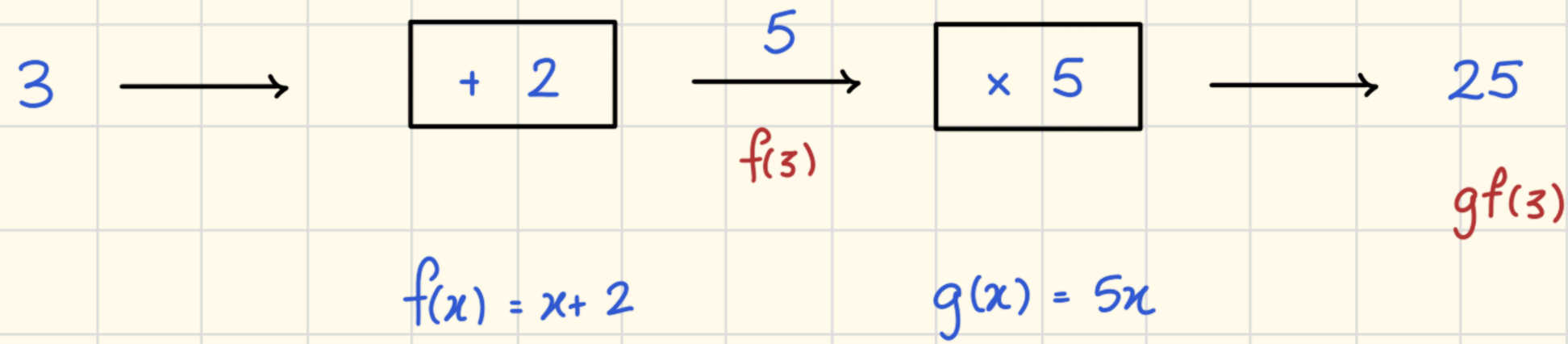
Math Error ⚠

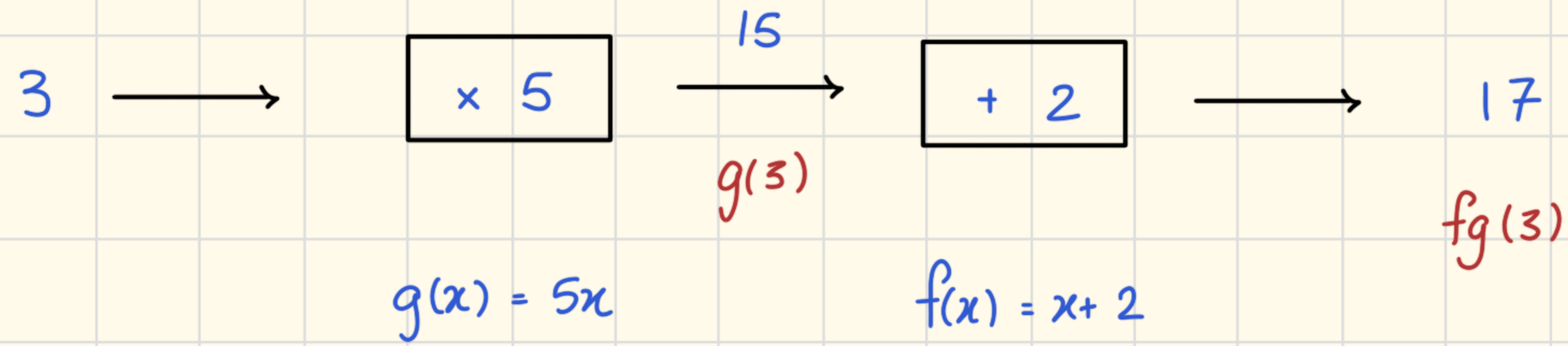
Reciprocal : $\frac{1}{x}$ $x \neq 0$

Square Root : $\sqrt{x}$ $x \not< 0$

FUNCTION COMPOSITION

A function composition is a combination of functions.





FUNCTION COMPOSITION

For $f(x) = 3x + 2$ and $g(x) = (x + 1)^2$, find

a) $fg(x) = f((x+1)^2) = 3(x+1)^2 + 2$

b) $gf(x) = g(3x+2) = (3x+2+1)^2$

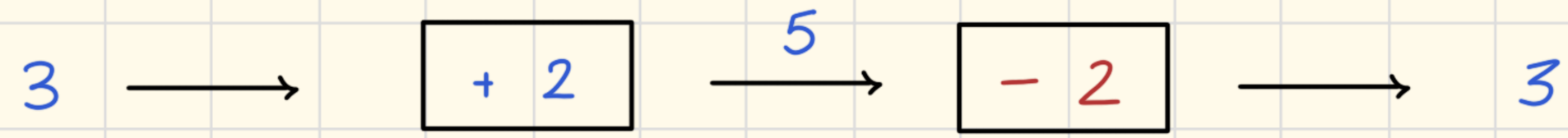
c) $ff(x) = f(3x+2) = 3(3x+2) + 2$

d) $gg(x) = g((x+1)^2) = ((x+1)^2 + 1)^2$

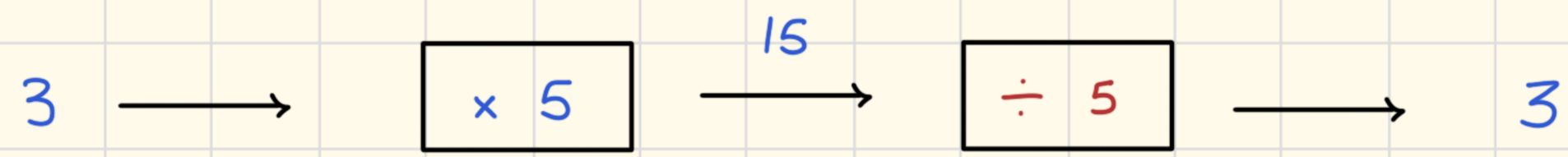
INVERSE FUNCTIONS

Some functions have *inverses* !

Inverse function go back to the original input .



$f(x) = x + 2$ $\quad$ $g(x) = x - 2$



$f(x) = 5x$ $\quad$ $g(x) = \frac{x}{5}$

INVERSE FUNCTIONS

You can find an inverse function by switching the x and y and solving for y !

Find inverse functions for :

$f(x) = x + 3$

$g(x) = 5x - 2$

$h(x) = (3x + 1)^2$

INVERSE FUNCTIONS

Find inverse functions for & Inverse

$f(x) = x + 3 \longrightarrow x = y + 3 \longrightarrow y = x - 3$

$g(x) = 5x - 2 \longrightarrow x = 5y - 2 \longrightarrow y = \frac{x+2}{5}$

$h(x) = (3x + 1)^2 \longrightarrow x = (3y+1)^2 \longrightarrow y = \frac{\sqrt{x} - 1}{3}$